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A digital twin-based quadruped robot system with scene perception, fast communication, and holographic interaction☆

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ABSTRACT

This paper develops a quadruped robot virtual-real interactive control system based on digital twin technology. The system is designed to address key challenges in robotics technology, including real-time performance, low-latency control, high-precision multi-sensor data fusion, stable network transmission, data security, user-friendly interaction interface, system scalability, and maintainability. The system comprises a number of functional modules, including a 3D modeling module, a positioning perception module, a virtual interaction module, a wise sensing-transmission module, and a cloud server. The 3D modeling module is responsible for constructing the virtual quadruped robot and motion space scenarios. The positioning perception module integrates LiDAR and Inertial Measurement Unit (IMU) data, utilizing Point-LIO and HDL-localization algorithms for high-precision environmental perception and positioning. The virtual interaction module provides a user-friendly control interface through computer software and the HoloLens headset. The wise sensing-transmission module employs WiFi and 5G links to ensure low-latency and high-bandwidth data transmission, and employs `libhv` and `libssl` asynchronous IO and network security cryptographic libraries to guarantee data security. The system is designed to run on the Ubuntu 20.04 platform, offering excellent scalability and maintainability. This system has broad application prospects in industrial manufacturing, construction, disaster rescue, military applications, and educational training. It enhances the performance and reliability of quadruped robot systems and lays a solid foundation for the future development of the industrial metaverse.

1. Introduction

In the current era, the rapid advancement of digitalization and intelligence is profoundly reshaping various industries [1,2], aligning with the transformative wave of Industry 4.0 which fosters smart manufacturing through the integration of digital technologies. This paradigm shift creates an urgent motivation for developing advanced virtual-real interactive systems that can operate robustly and efficiently, particularly for complex robotic platforms like quadruped robots in dynamic environments. Such systems are pivotal for realizing the full potential of the industrial metaverse, yet their practical, high-performance implementation remains a significant research endeavor. Robotics technology has become a key force in industrial manufacturing, the service sector, and daily life [3–5], serving as a cornerstone of the automation goals within Industry 4.0 [6]. As the core of the industrial metaverse [7–9], and a critical enabling technology for Industry 4.0, digital twin technology creates a precise digital simulation to construct a multi-dimensional dynamic virtual mirror of physical entities [10–12]. This provides new ways for the management, maintenance, and optimization of robotic systems.

The industrial landscape is undergoing a significant transformation with the integration of digital technologies and intelligent systems [13], a trend central to the Industrial Internet of Things (IIoT) [14] and the broader Industry 4.0 vision. Digital twins, which are virtual representations of physical entities, play a crucial role in this evolution by enabling data-driven insights and control in smart manufacturing

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environments. By creating a real-time, dynamic digital counterpart of a physical robot, digital twin technology enables comprehensive monitoring, analysis, and control [15]. This technology is particularly beneficial in the context of quadruped robots, which are increasingly used in diverse applications ranging from industrial automation to disaster response [16]. Effective interaction with such advanced Cyber-Physical Systems (CPSs) necessitates user-friendly Human-Machine Interfaces (HMIs), the evolution of which is critical for realizing human-centric systems, for example, in the context of Society 5.0 [17].

Despite the promising potential of digital twin technology, particularly its role in sophisticated Industry 4.0 applications and the IIoT, several challenges remain in developing an effective virtual-physical interaction system for quadruped robots. These challenges include: achieving real-time and low-latency control, realizing high-precision multi-sensor data fusion [18], ensuring the stability of network transmission and data security (especially vital in distributed IIoT architectures), improving the accuracy of panoramic small target detection and tracking algorithms, and designing user-friendly interfaces and a system with good scalability and maintainability.

Recent advancements have led to the development of various systems that aim to address these challenges. For instance, the implementation of Simultaneous Localization And Mapping (SLAM) algorithms [19,20] such as Point-LIO [21] and HDL-localization [22] has significantly improved the precision of environmental mapping and robot localization. Furthermore, advancements in wireless communication technologies [23,24], like 5G, have enhanced the reliability and speed of data transmission between virtual and physical entities.

Several studies have explored different aspects of digital twin technology and its application in robotics. Dormido et al. presented a scalable approach for CPS using ROS 2, which highlighted the importance of flexible frameworks for managing real and virtual robots [25]. This study emphasized the middleware capabilities of ROS 2 in ensuring efficient data exchange and control. The authors in [26] proposed a framework named Synchronous Mixed Reality (SMR) to personalize the virtual-real fusion system and maintain a balance between interaction efficiency and environmental immersion. This is achieved by combining an instance segmentation algorithm with an interaction prediction algorithm. In [27], a human interface framework of a CPS based on virtual reality, named the immersive and interactive CPS, is proposed to create an interface for human-machine-autonomy collaboration. The integration of heterogeneous physical systems with virtual assets, such as digital models, digital shadows, and virtual traces of human works, was achieved through the combination of a data-to-information protocol and middleware. The authors in [28] discussed an object-oriented event-driven simulation as a digital twin of a flexible assembly cell coordinated with a robot to perform assembly tasks alongside human. An approach to machine learning that enables industrial robots to navigate workspace safely around obstacles and people is presented in [29]. In [30], the authors investigate the use of a digital twin system to achieve virtual-physical tracking control for a carlike mobile robot. The digital twin system comprised a physical entity, a virtual entity, and a virtual–physical interaction module for the carlike mobile robot. Work [31] introduced TelePhysicalOperation, a teloperation concept that relies on a virtual physical interaction interface between the human operator and the remote robot. The proposed approach enables the user to virtually interact with the remote robot through the application of virtual forces, which are generated by the operator tracking system and can be selectively applied to any body part of the remote robot along its kinematic chain. In addition, some papers focus on quadruped robots. The authors in [32] proposed a digital twin-driven intelligent control system for robots to work in the home for stabilized people. The system is based on three pillars: data perception, digital twin construction, and human-machine interaction, which allows for the full fusion of robot virtual control with actual control. In order to monitor the real-time operation status of the robot and make necessary adjustments, [33] focused on the single-leg of a quadruped robot, proposes a single-leg virtual-real interaction system based on digital twin, and studies its virtual-real interaction characteristics. [34] presented mix reality for robotic assistance in post-disaster scenarios. The objective is to enhance the safety and effectiveness of rescuers by providing them with augmented reality guidance and environmental awareness. This technology enables robots to navigate challenging environments, such as collapsed structures, electrical hazards, and toxic substances, while also allowing them to perceive and process visual information about their surroundings.

These studies collectively advance the field by addressing critical challenges and providing innovative solutions for the integration of digital twins with quadruped robots, enhancing their capabilities in various applications.

However, while existing research has significantly advanced specific components of digital twin technology and robotics, many studies focus on isolated aspects or simulations. A clear gap remains in the availability of holistically implemented and validated virtual-real interaction systems that seamlessly integrate advanced perception, multi-modal communication, precise synchronization, and intuitive human control for highly dynamic robots like quadrupeds operating in real-world scenarios. This research is therefore motivated by the need to bridge this gap. This paper designs and implements a comprehensive virtual-physical interaction control system for quadruped robots based on digital twin technology. The aim is to achieve high efficiency, safety, and real-time control, creating a seamless bridge between the virtual and real worlds, thereby providing a new implementation pathway and innovative solutions for the forefront exploration of the industrial metaverse.

To address these challenges and fulfill the stated aims, this paper makes several key contributions. Firstly, we present the design, implementation, and validation of a complete and holistic digital twin-based virtual-real interaction system specifically for quadruped robots. A core domain-independent innovative advance of this system is its architectural framework for data management and communication, embodied in the wise sensing-transmission module. This module uniquely integrates WiFi and 5G technologies [35] with robust security protocols to ensure low-latency, high-bandwidth, and secure data flow between the physical robot, its digital twin, and remote operators. Furthermore, the system achieves high-precision environmental perception and robot localization by effectively fusing data from LiDAR and Inertial Measurement Unit (IMU) sensors using advanced SLAM algorithms like Point-LIO and HDL-localization, coupled with Extended Kalman Filter (EKF) [36] for state estimation. It also provides intuitive human-robot interaction through user-friendly interfaces developed for personal computers and HoloLens headsets. Finally, the practical efficacy of the proposed system is demonstrated through comprehensive tests, highlighting its real-time performance, operational stability, and inherent scalability and maintainability rooted in its modular design and the use of open-source platforms.

The key contributions are summarized as follows:

- **Real-time and Low-latency Control:** To ensure the robot can respond to remote control commands in real-time, the project optimizes communication protocols using wise sensing-transmission module. Through efficient WiFi and 5G links, data transmission latency is significantly reduced. Additionally, data processing algorithms are optimized to enhance the overall system's real-time performance and response speed.
- **High-precision Multi-sensor Data Fusion:** We employ Point-LIO and HDL-localization algorithms, integrating LiDAR and quadruped robots with depth cameras, LiDAR, and IMU data. By utilizing LiDAR-vision fusion algorithms, the system perceives the surrounding environment, and the EKF is used for state estimation, 3D mapping, real-time map updates, and obstacle avoidance. Efficient fusion and precise processing of multi-sensor data ensure the environmental perception and positioning accuracy of the quadruped robot.
- **Stable Network Transmission and Data Security:** By using the industrial-grade RK3588 motherboard, Quectel RM500 5G module,

and MT7921K WiFi 6E module, the system optimizes network transmission stability. Security libraries such as `libhv` and `libssl` are introduced to ensure the security and integrity of data transmission, preventing data tampering or leakage during remote control.

- **Panoramic Small Target Detection Algorithm:** The system introduces the ResNet network [37,38] for feature extraction, combined with the Spatial Pyramid Pooling (SPP) structure [39] and spatial and pixel attention mechanisms, improving target detection accuracy.
- **User-friendly Interface:** Personal Computer (PC) software and HoloLens [40] headset interfaces are developed, allowing users to control virtual and real robots through an intuitive interface. Real-time feedback and correction mechanisms ensure consistency between the virtual and real robot states, enhancing user experience and control accuracy.
- **System Scalability and Maintainability:** The system's choice of Ubuntu 20.04, a widely-used open-source long-term support Linux distribution, is pivotal for its scalability and maintainability. Ubuntu's open-source nature provides access to an extensive ecosystem of software packages, libraries, and development tools, simplifying the integration of new functionalities and hardware, thus facilitating system expansion. This includes robust support for asynchronous I/O operations and network security encryption libraries. Furthermore, strong community and commercial support for Ubuntu ensures readily available documentation, troubleshooting resources, and regular security updates, which are crucial for long-term maintainability and stable operation. The Linux environment also inherently supports the system's modular design, allowing individual components and algorithms to be updated or replaced with minimal impact on the overall system. Coupled with broad hardware compatibility beneficial for integrating diverse robotic sensors and peripherals, the use of Ubuntu 20.04 provides efficient maintenance, robust expansion capabilities, and ensures long-term stable operation of the quadruped robot system.

The system includes five parts: 3D modeling module, wise sensing-transmission module, virtual end interaction module, positioning perception module, and cloud server. The output of the 3D modeling module is connected to the virtual end interaction module; the virtual end interaction module is connected to the input of the cloud server; the wise sensing-transmission module connects the cloud server and the positioning perception module, facilitating information exchange between the cloud server and the physical entity of the quadruped robot. The cloud database is part of the cloud server, used to store positioning information, perception information, scene information, and control instruction data. This system can be applied to various scenarios such as environmental intelligent perception, HoloLens module development, quadruped robot and training scene integration development, and virtual-physical interaction of quadruped robots. In these services, user control data are sent to the quadruped robot via the cloud server, and the quadruped robot's collected state data is fed back to the cloud server to correct the virtual robot, achieving information communication, model fusion, and virtual-physical interaction between the real and virtual quadruped robots.

The remainder of this paper is organized as follows. The system model and the framework of the digital twin-based quadruped robot system are described in Section 2. The hardware design is presented in Section 3. The software design is presented in Section 4. System tests are analyzed in Section 5. The discussion of potential applications is presented in Section 6. Conclusions are drawn in Section 7.

2. System model

The framework of the digital twin-based quadruped robot virtual-reality mutual control system mainly consists of five parts, including the quadruped robot physical twin, virtual twin, twin data, service, and the interconnections between each component, as shown in Fig. 1.

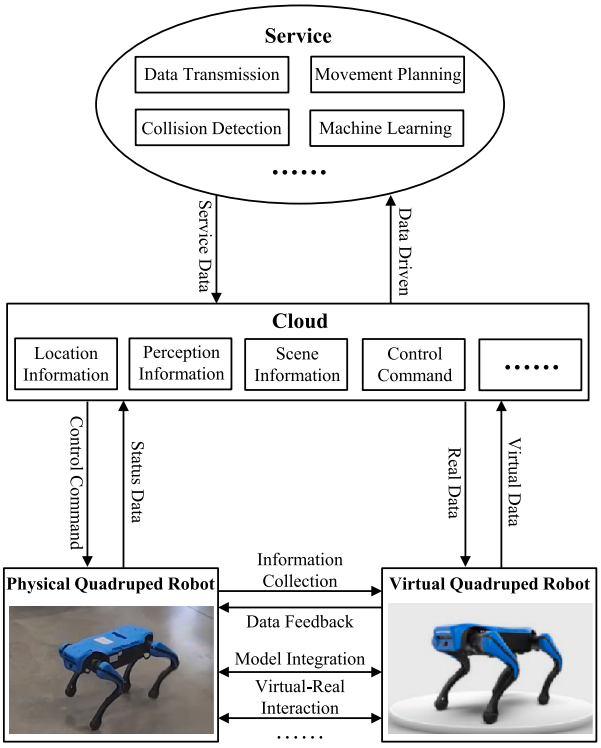


Fig. 1. The framework of the digital twin-based quadruped robot.

2.1. Physical twin

In the quadruped robot virtual-physical interaction system, the physical twin refers to the quadruped robot in the real world. Its primary responsibilities are to execute specific actions, interact with the environment, and feedback execution results to the virtual twin. The physical twin mainly consists of the following components:

- **LiDAR:** Used to measure the distance and position information of the surrounding environment, enabling environmental perception and obstacle detection.
- **Depth Camera:** Acquires depth information of objects in the environment using infrared technology, allowing for 3D reconstruction and object recognition.
- **Wide-angle Camera:** Captures images and videos of the environment, providing visual information for the robot system to process and make judgments.
- **External Interfaces:** Used to connect with other devices or sensors, such as controllers and computers, to achieve data exchange and communication.
- **Legs (Thigh and Shank):** These are the main movement components of the quadruped robot, with joints and actuators controlling the robot's gait and posture.
- **Anti-slip Foot Pads:** Designed with anti-slip materials or structures to enhance the robot's stability and grip on different surfaces.
- **Wise Sensing-Transmission Equipment:** The wise sensing-transmission node acts as the data hub between the quadruped robot and the server. It consists of an industrial-grade RK3588 motherboard, a Quectel RM500 5G module, an MT7921K WiFi 6E module, and a 19,200 mAh high-capacity battery. It is responsible for transmitting sensor data and execution results collected by the quadruped robot to the server and receiving control commands from the server.

2.2. Virtual twin

The virtual twin refers to the quadruped robot created in the virtual reality environment. Its primary responsibilities are to interact with various objects within the virtual environment. For instance, it can communicate and collaborate with scenes and other virtual entities within the virtual environment. Additionally, the virtual twin needs to receive data transmitted by the physical twin, such as position, speed, and actions, and adjust its state accordingly to maintain synchronization with the physical twin.

2.3. Twin data

Twin data is used to record and store the data exchange and synchronization information between the physical twin and the virtual twin. This includes sensor data from the physical twin as well as the state, position, and actions of the virtual twin. By establishing a one-to-one mapping relationship between the physical and virtual entities through twin data, virtual-physical interaction control can be achieved.

2.4. Service

The service module provides various functions and algorithm support for processing data, control commands, and algorithm computations for both the physical and virtual twins. Specifically, the service module offers comprehensive support and capabilities for virtual-physical interaction control through real-time data transmission, motion planning, collision detection, and machine learning functions.

2.5. Interconnections between each component

The interconnection module integrates the physical twin, virtual twin, twin data, and service module through various information interfaces and communication protocols, enabling data exchange and transmission of control commands [41,42]. The connection is bidirectional, allowing the physical and virtual twins to perceive, interact, and synchronize with each other, thus achieving a closed-loop system for virtual-physical interaction.

2.6. Workflow

Based on the aforementioned system framework, the workflow of the system is as follows:

1. **Data Collection:** LiDAR and IMU continuously collect environmental and motion posture data, transmitting it to the positioning and perception module.
2. **Data Processing:** The positioning and perception module combines multi-sensor data, using Point-LIO and HDL-localization algorithms for data fusion and positioning calculations, and employs EKF for state estimation.
3. **Data Transmission:** The processed positioning and state data are transmitted to the local perception host via the WiFi 6E module and uploaded to the cloud server via the 5G module.
4. **Remote Control:** The cloud server receives user control commands and sends them through the 5G link to the wise sensing-transmission node, which then transmits the control commands to the motion host via the WiFi link, driving the quadruped robot to perform the corresponding actions.
5. **Feedback Correction:** After executing actions, the quadruped robot feeds back real-time position and state information to the virtual end interaction module, which corrects the virtual robot's state, ensuring synchronization between the physical and virtual twins.

Through the above design scheme, this system achieves efficient, stable, and real-time virtual-physical interaction control for the quadruped robot, enhancing the overall performance and reliability of the system.

3. Hardware design

The design and implementation of the hardware components are crucial to ensuring the efficient operation of the quadruped robot virtual-physical interaction control system.

3.1. RK3588 Industrial-grade motherboard

The RK3588 industrial-grade motherboard serves as the main control and processing unit of the system, responsible for computationally intensive tasks including data processing and control. This System on a Chip (SoC) typically features an octa-core CPU (e.g., Quad-core ARM Cortex-A76 and Quad-core Cortex-A55), a Mali-G610 GPU for graphics acceleration, and a dedicated Neural Processing Unit (NPU) supporting AI computations up to 6 TOPS. Such motherboards are often complemented by configurable LPDDR4/LPDDR4X/LPDDR5 RAM options, commonly ranging from 4 GB to 16 GB, or even 32 GB in some configurations. Its high-performance processing capabilities across general-purpose, graphics, and AI tasks, along with rich interface support, provide the system with robust computational resources and flexible expansion capabilities crucial for the demanding applications of the quadruped robot.

3.2. Quectel RM500 5G module

The Quectel RM500 5G module is used for high-speed data transmission. The introduction of 5G technology significantly increases data transmission bandwidth and stability, ensuring low latency and high reliability in remote control and real-time feedback. The 5G link establishes TCP Socket and WebSocket connections with the cloud server to upload sensor data and transmit control commands.

3.3. MT7921K wifi 6E module

The MT7921K WiFi 6E module is used for high-speed data transmission within the local area network. WiFi 6E technology provides higher transmission rates and lower latency, ensuring efficient communication for the quadruped robot in local environments. The WiFi link connects to the access point of the quadruped robot, responsible for UDP data transmission between the perception host and the motion host.

3.4. Lidar and IMU

LiDAR and IMU are the core sensors of the positioning and perception module. LiDAR acquires distance information of objects in the environment by emitting laser beams and measuring the time it takes for the reflections to return, forming high-precision point cloud data. The IMU provides motion posture data. The data from both sensors are processed using the Point-LIO and HDL-localization algorithms, enabling high-precision positioning and environmental perception for the robot.

These components collectively ensure the quadruped robot system operates efficiently, supporting the high-performance requirements of virtual-physical interaction control.

4. Software design

The software modules are the soul of the quadruped robot virtual-physical interaction control system.

4.1. 3D Modeling module

Scene construction plays a crucial role as the primary task in the development of the quadruped robot digital twin system. This paper uses VeryEngine as the development platform, which includes various functional modules such as model editing, rendering editing, and user interface editing. The flowchart of 3D scene construction is shown in Fig. 2.

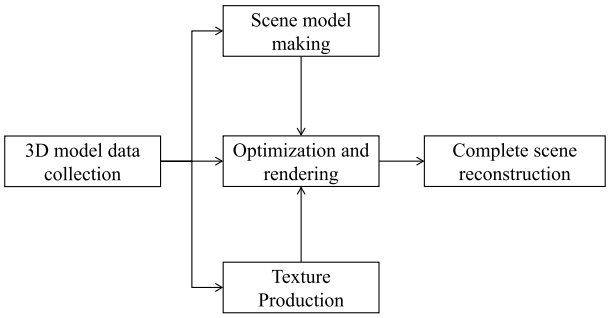


Fig. 2. Flowchart of 3D scene construction.

The 3D reconstruction process is divided into the following four aspects:

- We employ Physically Based Rendering (PBR) [43] to create a realistic texture model of the quadruped robot twin, with dimensions identical to the actual device.
- We capture the real-world buildings for experiment, including the public environment configuration and general facilities and equipment in the test building.
- We use the onboard LiDAR of the quadruped robot to pre-model the deployment scene globally. By emitting and receiving laser beams to obtain time differences, the system calculates distance and signal intensity information, resulting in point cloud data and object surface characteristics, such as color or material.
- The point cloud data is filtered and converted into points in a 3D coordinate system. The Point-LIO algorithm is used to incrementally build the 3D structure of the environment.

The Point-LIO algorithm is based on two core principles: LiDAR point cloud sampling in time sequence and IMU data as system measurements. LiDAR points and IMU data are used to update the state at each timestamp, achieving a high-frequency state output of 4–8 kHz. The detailed process is as follows:

4.1.1. Lidar point processing

- For each received LiDAR point, the corresponding plane is searched within the existing map.
- If plane fitting is achieved, residuals are calculated to update the pose, and the LiDAR point is registered into the global framework and merged into the map.
- If no matching plane is found, the pose estimated using the EKF filter is directly added.

4.1.2. Fast plane search

The ikd-Tree structure [44] is used to achieve fast plane correspondence search.

4.1.3. IMU Data processing

Each IMU measurement is individually checked for saturation in each channel, and saturated channels are not used for state updates.

In the context of this system, the LiDAR point cloud data provides observations in terms of radial distance and angle, which form a nonlinear relationship with the robot's state. Unlike the basic Kalman Filter, which assumes linear models, the EKF allows for nonlinear state transition and observation models [45]. These nonlinear models can be linearized around the current estimate using a Taylor series expansion. The EKF employs Jacobian matrices to linearize the nonlinear state transition and observation functions. The Jacobians are computed at each time step based on the current-state estimate, and they provide a linear approximation of the nonlinear functions.

Here is how the EKF works in the system [46].

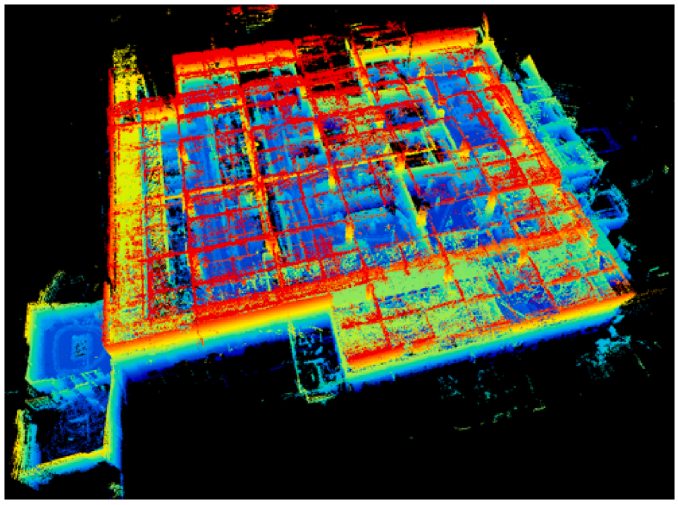


Fig. 3. Illustration of the modeled 3D map.

Predict the state:

$$\hat{x}_{k|k-1} = f(\hat{x}_{k-1|k-1}, u_{k-1}) \tag{1}$$

Predict the covariance:

$$P_{k|k-1} = F_{k-1} P_{k-1|k-1} F_{k-1}^T + Q_{k-1} \tag{2}$$

Compute the Kalman gain:

$$K_k = P_{k|k-1} H_k^T (H_k P_{k|k-1} H_k^T + R_k)^{-1} \tag{3}$$

Update the state estimate:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k(z_k - h(\hat{x}_{k|k-1})) \tag{4}$$

Update the covariance:

$$P_{k|k} = (I - K_k H_k) P_{k|k-1} \tag{5}$$

Here, f is the nonlinear state transition function, F_{k-1} is the Jacobian of f with respect to the state, Q_{k-1} is the process noise covariance, h is the nonlinear observation function, H_k is the Jacobian of h with respect to the state, and R_k is the measurement noise covariance.

By using the EKF with Jacobian matrices, the system can effectively handle the non-linearities present in the LiDAR point cloud data, leading to accurate state estimation and improved performance of the quadruped robot digital twin system.

The specific modeled 3D map is illustrated in Fig. 3.

4.2. Position perception module

The position perception module is the core of the virtual-physical interaction control system for the quadruped robot. This module is responsible for target localization and intelligent perception. In this system, the quadruped robot functions as an execution terminal within the virtual-physical interaction system, receiving remote control commands and transmitting information such as coordinates, attitude angles, and robot status. The localization information is combined with the modeling map, IMU, and high-angle panoramic camera for map updates and target detection. The processing flow is shown in Fig. 4.

4.2.1. Target localization and intelligent perception

The system achieves target localization and intelligent perception by jointly deploying the Point-LIO mapping algorithm and the HDL-localization positioning algorithm. The algorithm framework is illustrated in Fig. 5. The HDL-localization algorithm is based on the Unscented Kalman Filter (UKF) [47] framework and integrates the Normal Distributions Transform (NDT) [48] point cloud registration results to

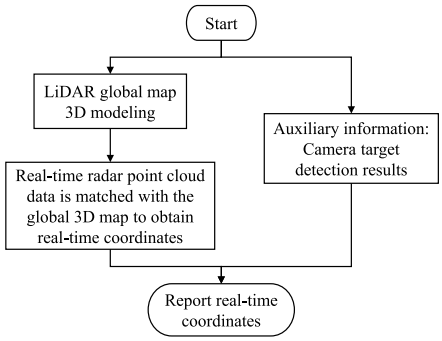


Fig. 4. Flowchart of the localization information processing algorithm.

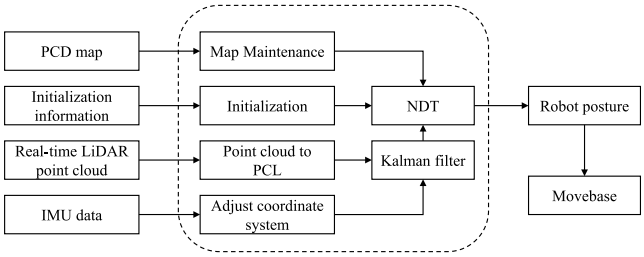


Fig. 5. Block diagram of global map 3D modeling generation and point cloud data matching algorithm.

achieve laser-based relocalization. The specific operations are as follows:

Data Acquisition and Preprocessing

- Use LiDAR and odometry data to model the 3D scene.
- Grid the point cloud data, dividing it into specified-sized voxels.

NDT Algorithm Registration

- Calculate the multidimensional normal distribution parameters (mean and covariance matrix) for each voxel.
- Initialize the transformation parameters.
- Transform the point cloud to be registered to the reference point cloud's grid.
- Calculate the probability density of each transformed point based on the normal distribution parameters.
- Sum the probability densities calculated for each voxel to obtain the NDT registration score.

Optimization

- Use the Newton optimization algorithm to optimize the NDT registration score, finding the optimal transformation parameters that maximize the registration score.
- Calculate the gradient and Hessian matrix of the objective function during optimization.

3D Point Cloud Matching

- Describe the rotation using 3D Euler angles and optimize six transformation parameters (three translations and three rotations).

In summary, the quadruped robot ensures its capability as an execution terminal by combining the SLAM mapping algorithm Point-LIO and the HDL-localization positioning algorithm. Utilizing multi-sensor data fusion and efficient matching algorithms, the robot can receive control commands from the server and transmit position information (3D spatial coordinates, and attitude angles) and status information (walking, running, turning, speed, acceleration) through the wise sensing-transmission node to the server. This data is used for subsequent processing and decision making.

4.2.2. Target detection and tracking

The target detection and tracking module provides the quadruped robot with auxiliary localization information and spatial position frames, thereby enhancing the precision of the localization algorithm. A panoramic camera mounted at a high point is primarily employed to document the experimental setting and obtain uninterrupted video footage of the panoramic image. However, the panoramic image possesses a considerable depth of field, and the quadrupedal robot occupies a relatively minor proportion at the far end of the frame (typically less than 32×32 pixels), which lacks sufficient visual information to differentiate it from the background or analogous targets. Furthermore, the experimental scene is intricate and complex, and is typically subject to significant fluctuations in illumination, target occlusion, the presence of multiple closely-spaced targets, and changes in target scale. These factors collectively contribute to the inherent difficulty of detecting small targets.

To enhance the precision and resilience of quadruped robot detection in complex scenes with a vast depth of field while maintaining the real-time operational efficiency of the system, this paper proposes a target detection and tracking network structure comprising a target detection module, a motion detection module, and a trajectory tracking module. The target detection module is responsible for introducing an attention mechanism to perform preliminary target detection on the panoramic image. The motion detection module is designed to complement the expansion of the target detection module, thereby enhancing the recall rate of target detection. The trajectory tracking module employs the optical flow method to obtain the trajectory information of the target between video frames. This information is then processed by a post-processing network to filter out targets that were omitted or misdetected by the target detection module.

Target Detection Module

The target detection module first preprocesses the panoramic images. To avoid losing image details during scaling, the panoramic image is divided into nine regions and fed into the network for detection. The backbone network uses ResNet-50 for image feature extraction, followed by SPP to retain both local and global textual information, as shown in Fig. 6.

SPP obtains pooled feature maps, and a channel attention mechanism is introduced. This mechanism involves pooling the features, using a fully connected network and an activation function to get a weight matrix that assigns more weight to channels containing the target. The weighted enhanced feature map is obtained by multiplying the weight matrix with the feature map. Similarly, the pixel attention mechanism assigns more weight to regions containing the target within the feature map, and the weighted mask is multiplied with the channel attention-enhanced feature map to obtain preliminary target detection results.

Motion Detection Module

While the attention mechanism helps improve the recall rate of the target detection module for small targets, missed detections can still occur. The motion detection module, therefore, is introduced to improve robustness. The ViBe algorithm [49] is used to extract the background, establishing a sample set for all pixels, storing past pixel values and neighboring pixel values. The current pixel value is compared with historical values to determine if a pixel belongs to the background.

Tracking Detection Module

Post-integration of target and motion detection results, considering inter-frame video correlations, the tracking detection module employs motion gradient methods to identify potential target areas. The design is shown in Fig. 7.

After obtaining three consecutive frames, target center point information is used to calculate forward and backward optical flow. Maximum motion gradient is calculated as $G = \max\left\{\sqrt{(u_x+u_y)^2}, \sqrt{(v_x+v_y)^2}\right\}$, where u_x, u_y, v_x, v_y are vector speeds on the x and y axes. $M = \max\{G_{0-1}, G_{1-2}, G_{2-1}, G_{1-0}\}$ represents the maximum motion gradient of the optical flow between the three

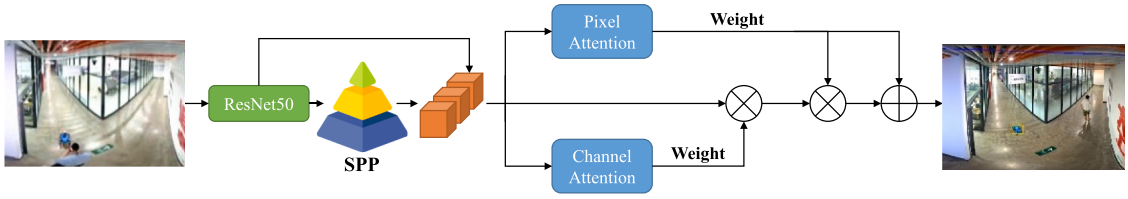


Fig. 6. Simplified framework of target detection.

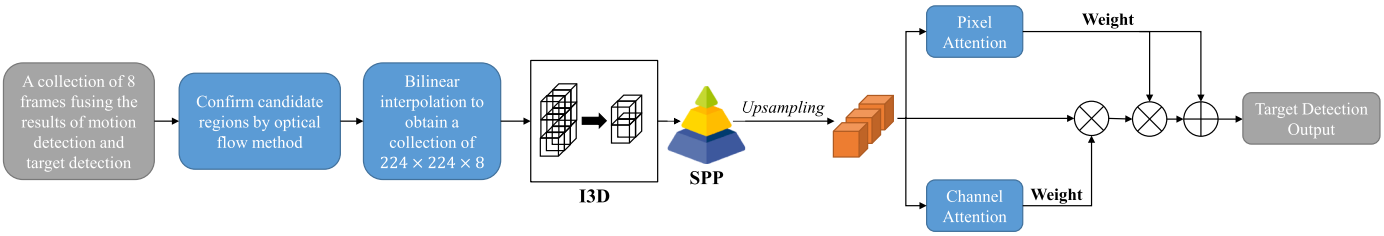


Fig. 7. Design of the tracking detection module.



Fig. 8. Illustration of the target detection result.

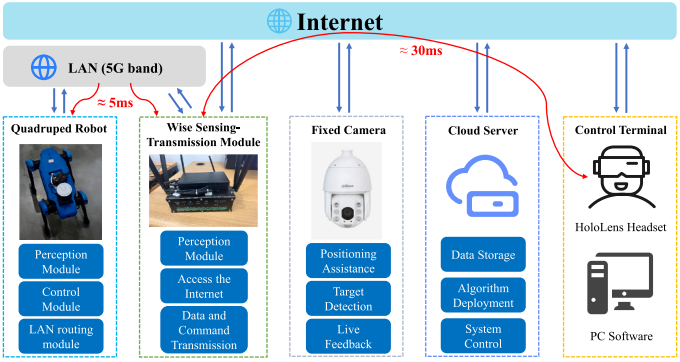


Fig. 9. Block diagram of digital twin-based virtual-real control system for quadruped robots.

frames. In some cases, the optical flow method may fail to identify all the candidate pixel points. To address this issue, the method employs a conditional random field to enhance the correlation between neighboring pixel points and expand the potential target area.

Potential target areas from three frames generate small images of 100×100 pixels, forming a $100 \times 100 \times 8$ set over eight frames. Bilinear interpolation scales it to $224 \times 224 \times 8$, feeding into the tracking network. The backbone network uses an improved Inflated 3D ConvNet (I3D) for temporal feature extraction. The benefit of I3D lies in its pre-trained GoogleNet backbone, operating as a dual-stream network performing 3D convolutions on RGB images and optical flow, ensuring fast operation and low memory usage, achieving state-of-the-art performance. Optionally, the base network can be switched to Inception-v4, decomposing 5×5 and $n \times n$ convolutions to reduce parameters and introducing residual connections to prevent gradient vanishing and reduce model complexity.

Post I3D feature extraction, the channel and pixel attention mechanisms perform secondary detection to obtain final panoramic target detection results, as shown in Fig. 8.

4.3. Virtual interaction module

The virtual interaction module is used to transmit user control commands to both virtual and physical quadruped robots. The real-world quadruped robot feeds back the position and status information obtained by the localization and perception module to correct the virtual quadruped robot, enabling mutual control between virtual and physical devices. The specific method for transmitting user control commands to the virtual and real quadruped robots is as follows:

- The user inputs control commands via mouse and keyboard operations on the PC software or through the controller on the HoloLens headset.
- The virtual quadruped robot performs corresponding actions based on the commands.
- The wise sensing-transmission node hardware issues commands to drive the real quadruped robot.
- The physical quadruped robot feeds back position and status information, and the wise sensing-transmission node hardware uploads this information to correct the virtual quadruped robot.

The interaction system for virtual-real control is illustrated in Fig. 9. The wise sensing-transmission module serves as a relay node, establishing a connection between the Local Area Network (LAN) and the Internet of the physical and virtual quadruped robot. It transmits both the sensing information in the actual scene and the control instructions of the virtual terminal, thereby facilitating virtual-real control. A trial run of the system revealed that the time delay for transmitting control signals from the control terminal to the actual scene is approximately 35 ms, which comprises a 5-ms delay from the wise sensing-transmission module to the quadruped robot via the 5G LAN link and a 30 ms delay from the control terminal to the wise sensing-transmission module through the Internet.

4.4. Wise sensing-Transmission module

The wise sensing-transmission module is used to send server control commands to the local quadruped robot via remote transmission,

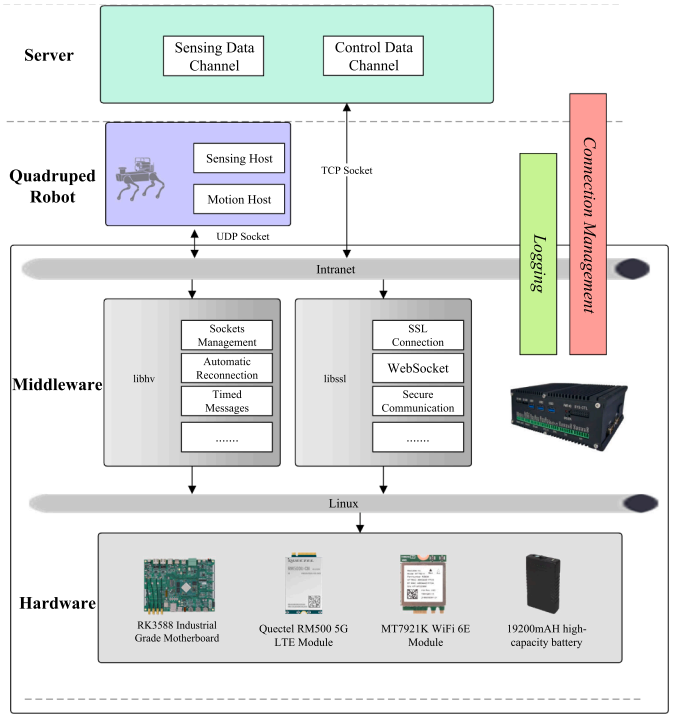


Fig. 10. Block diagram of the wise sensing-transmission module in the quadruped robot virtual-physical interaction system.

allowing remote control of the quadruped robot. To ensure the security and integrity of data transmission, the system employs SSL connections, implemented using the `libssl` library, alongside other security measures. The specific framework of the wise sensing-transmission module for the virtual-real interaction system is shown in Fig. 10, in which:

- The WiFi link connects to the quadruped robot's access point and is responsible for UDP Socket data transmission between the perception host and the motion host.
- The 5G link establishes TCP Socket and WebSocket connections with the server, transmitting perception host data to the server and receiving motion control messages from the server. The IP of devices under the 5G link is assigned by the carrier network through DHCP.

5. System tests and analysis

In the developed data video stream visualization interface, Fig. 11 shows the perspective seen by the user through the VR headset. Users can input control commands via mouse and keyboard operations on the PC software or through the controller on the HoloLens headset. Using the buttons on the control interface, the virtual quadruped robot performs the corresponding actions based on the commands. The following are the recommended hardware and software environment requirements for running the PC application. The system should be equipped with an Intel i5 processor, 8 GB of RAM, and 1 TB of hard drive space. Network requirements include a 1000 Mbps Ethernet card and an internet speed of 2 MB/s. Additionally, the setup requires at least one USB 3.0 port and a video output of DisplayPort 1.2 or a newer version. The software is compatible with Windows 7 or any later operating system. The wise sensing-transmission hardware issues these commands to drive the real quadruped robot. Through this system, participants can freely roam in the virtual scene and operate the robot according to task prompts, as shown in Fig. 12.

The system's capability for real-time interaction and low-latency control is fundamentally supported by its communication architecture. The



Fig. 11. Perspective seen by the user through the HoloLens headset.

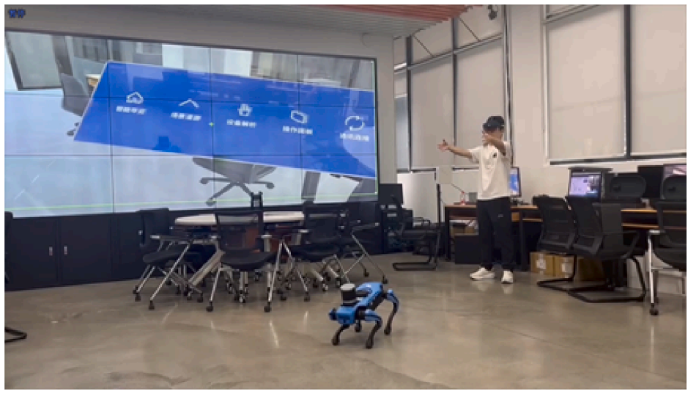


Fig. 12. Illustration of the use of the quadruped robot virtual-physical interaction system.

communication system is designed for low latency, with observed end-to-end control signal delays from the control terminal to the actual scene being approximately 35 ms. A summary of the key communication links, technologies, and protocols employed is provided in Table 1.

Fig. 13 illustrates the spatial relationship between the physical and virtual quadruped robots at different moments. It can be observed that the positions of the physical and virtual quadruped robots are closely aligned at the same moments, which validates the precision and low latency of the designed virtual-real interaction system. The movements of the quadruped can be controlled by manipulating the PC software or HoloLens headset software, and the status of the quadruped can be viewed in real time within the software.

The errors between the virtual twin and the physical twin of our simulated quadruped robot system in the X, Y coordinates, and angles are shown in Fig. 14. It can be seen that the errors are relatively small, mostly within ±1 meter and ±0.4 degrees.

As shown in Fig. 15, we simulated the distribution of actual coordinates (Real points) and virtual coordinates (Virtual points) on the X and Y coordinate plane under four different angles. It can be seen that the distribution of virtual coordinates is quite consistent with the actual coordinates at most angles, demonstrating high positioning accuracy of the system under different angles. However, at certain specific angles (e.g. 18.2 degrees), there is still some error, indicating the need for further system optimization to improve overall positioning accuracy.

Table 1
Communication Links and Protocols in the Quadruped Robot System.

Link Name	Technology	Protocol
Robot to Cloud Server (Uplink/Downlink)	5G (Quectel RM500 module)	TCP, WebSocket
Robot Internal Communication (Local)	WiFi 6E (MT7921K module)	UDP Socket
Control Terminal to Wise Sensing-Transmission Module	Internet	TCP/IP (inferred)

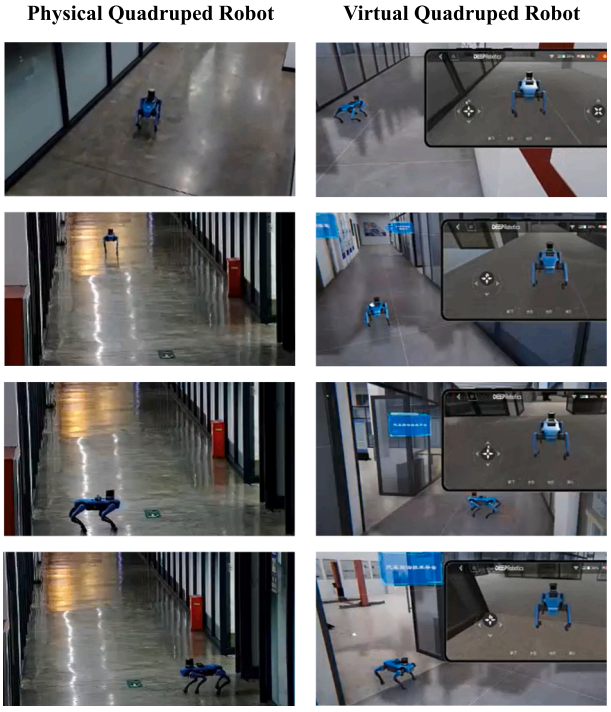


Fig. 13. Corresponding positions of physical and virtual quadruped robots at different moments.

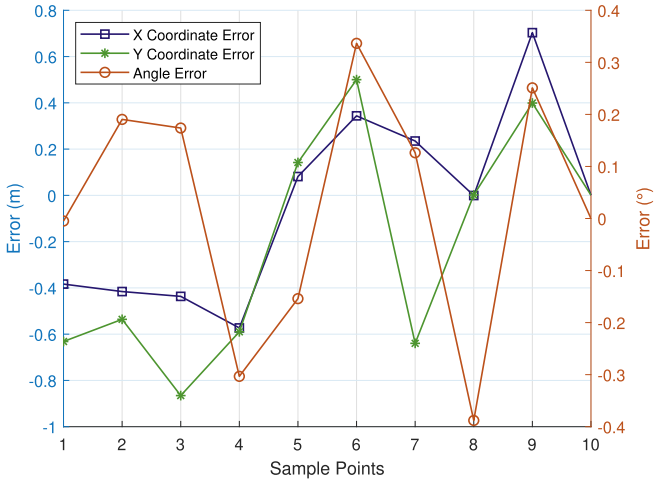


Fig. 14. X, Y coordinate and angle error.

6. Discussion

The virtual-physical interaction system for quadruped robots based on digital twins has broad application value, mainly reflected in the following aspects:

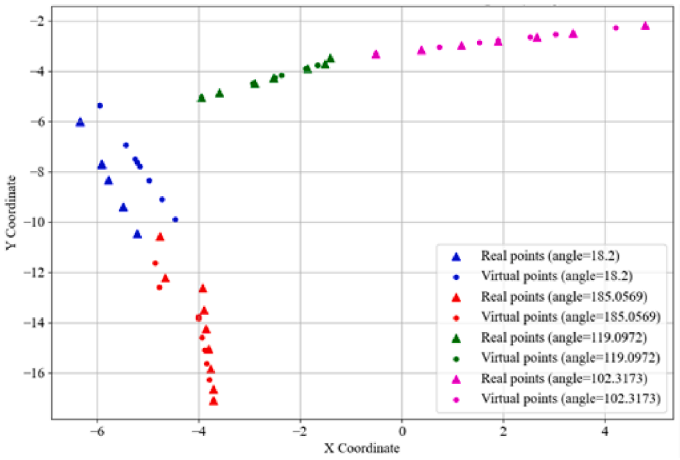


Fig. 15. Distribution of real and virtual coordinate points on the X and Y coordinate planes at different angles.

1. Remote Control and Maintenance

- Enables remote control and monitoring of both virtual and real quadruped robots, suitable for hazardous environments or remote areas.
- Participants can perform simulation testing, fault diagnosis, and parameter adjustment in a virtual environment. Afterward, control commands can be issued to the real robot, reducing on-site operational risks and costs.

2. Immersive Interactive Experience

- Provides an immersive quadruped robot control experience, where users can control the robot in a virtual environment and observe its motion state, suitable for educational training and entertainment applications.

3. Data Collection and Analysis

- Real-time acquisition and transmission of robot position data to the cloud for visualization and optimization of control algorithms, enhancing motion performance and adaptability.

4. Practical Application Scenarios

- **Industrial Manufacturing**: Optimize production processes, remote monitoring, and fault diagnosis.
- **Construction**: Plan operation paths in advance and remotely coordinate construction activities.
- **Disaster Rescue**: Remotely control robots to execute search and rescue missions, improving safety.
- **Military Applications**: Pre-plan tactical scenarios and remotely control reconnaissance robots.
- **Educational Training**: Provide a control training platform applicable to actual robot operation.

Overall, the virtual-physical interaction system for quadruped robots integrates virtual simulation and real equipment to enhance efficiency

and safety. It finds widespread application in industries such as manufacturing, construction, rescue, and military, laying a solid foundation for the future development of the industrial metaverse. As technology advances, this mode of virtual-physical integration will become a key direction for industrial transformation [50,51].

While the system showcases significant capabilities and broad application potential as outlined above, some limitations should be noted. The initial setup cost, driven by specialized hardware such as the quadruped robot itself, the RK3588 industrial-grade motherboard, 5G modules like the Quectel RM500, WiFi 6E modules like the MT7921K, advanced LiDAR and IMU sensors, and potentially software licensing (e.g., for 3D modeling tools like VeryEngine), may be considerable. Furthermore, while user interfaces for PC and HoloLens are implemented, a formal usability evaluation could provide valuable feedback for enhancing user experience. Future work could address these aspects by exploring cost-effective hardware alternatives and conducting detailed usability studies to further refine the human-machine interaction.

7. Conclusion

This paper designs and implements a virtual-physical interaction system for quadruped robots based on digital twins, encompassing the following modules: a 3D modeling module, position perception module, virtual interaction module, wise sensing-transmission module, and cloud server. The 3D modeling module constructs the 3D model of the virtual quadruped robot and its environment. The position perception module acquires real-time coordinates of the real quadruped robot through multi-sensor fusion and efficient matching algorithms. The virtual interaction module transmits user control commands to both the virtual and real quadruped robots, enabling real-time interaction. The wise sensing-transmission module remotely sends control commands from the server to the local quadruped robot. The cloud server acts as the central information hub, receiving user commands and issuing them to the virtual and real quadruped robots through the wise sensing-transmission module. Based on the digital twin approach, this system successfully realizes mutual control between remote virtual devices and real devices, demonstrating extensive application prospects in the context of the industrial metaverse.

For future research, we propose several directions to extend this work. These include enhancing the system's scalability to accommodate more complex environments, integrating advanced artificial intelligence for autonomous decision-making, and exploring its applicability in fields such as search and rescue or environmental monitoring. These avenues will further unlock the potential of digital twin technology in robotics.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRediT authorship contribution statement

Chen Zhu: Writing – original draft, Validation, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization; **Jianrong Bao:** Writing – review & editing, Supervision, Resources; **Zhouxiang Zhao:** Writing – original draft, Visualization; **Zhaohui Yang:** Writing – review & editing, Supervision, Resources, Project administration; **Chongwen Huang:** Writing – review & editing, Supervision, Resources; **Jiawen Kang:** Writing – review & editing, Resources; **Hao Xu:** Writing – review & editing, Resources; **Zhaoyang Zhang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

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